Solving Problems

Effectively collaborating with Ai

Danielle Presgraves, Ph.D.
Director, Data Science Training

The Jackson Laboratory

Day 2 Architecture of Reasoning

How do we expose hidden assumptions?
How do we break-down complex problems?

Workshop Overview: *Why do smart people make predictable mistakes?*

Day 1- What is Knowledge?

- Discussion: the crisis of authority and need for metacognition
- What is knowledge?
- Kahneman systems
- Intro to 4 frameworks

Day 2 – What is Reasoning?

- Discussion: Facione Paper
- Structure of an Argument
- Paul-Elder Framework
- Intellectual Autopsy
- Polya's 4 steps

Day 3 – Data Literacy

- Discussion: Dataset
- Epistemology of Data
- Wild & Pfannkuch PPDAC Cycle
- Synthesis of the four frameworks
- Turing Audit

The Calibration Problem: How do we know when to trust our own conclusions?

The Four Frameworks

Kahneman

System 1 - *FAST*

- Uses heuristics

System 2 – *SLOW*

- Deliberate, and analytical
- Cognitively demanding

Day 1

Paul-Elder

1. Purpose
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Polya

1. Understand Problem
2. Devise a Plan
3. Execute Plan
4. Iteration & Reflection

Day 2

Wild & Pfannkuch

1. Problem
2. Plan
3. Data
4. Analysis
5. Conclusion

Day 3

The Four Frameworks

Kahneman

System 1 - *FAST*

- Uses heuristics

System 2 – *SLOW*

- Deliberate, and analytical
- Cognitively demanding

Paul-Elder

1. Purpose
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Polya

1. Understand Problem
2. Devise a Plan
3. Execute Plan
4. Iteration & Reflection

Wild & Pfannkuch

1. Problem
2. Plan
3. Data
4. Analysis
5. Conclusion

Calibration failures these model's address:

We are unaware that fast thinking is operating at all!

We argue from hidden assumptions that we never examine!

We treat complex problems as unsolvable wholes!

We trust data that encodes errors that we cannot see!

Thinker’s Log

Entry 2

Thinker’s Log Entry #2

Prompt 1: What did I believe confidently this week that turned out to be wrong or uncertain?

Prompt 2: What evidence would change your most constant belief?

Yesterday's Homework

Discussion

Discussion: The Crisis of Authority

1. Facione Paper: what is “offensive violence”?
2. Mini-case study, Day 1
3. Personal Heuristic Protocol, Day 1

Structure of an Argument & Informal Fallacies

Lecture

What is an argument?

- Reasoning is the process of using evidence to support conclusions
- **Three broad categories:** we use **induction** and **abduction** to build hypotheses from evidence, **deduction** to test the logical consequences of those hypotheses

Deductive Reasoning	**Inductive Reasoning**	**Abductive Reasoning**
Starts with a **general rule** and applies it to a **specific case**. (top down)	Starts with **specific observations** and infers a **general pattern**. (bottom up)	Starts with **observations** and **infers** most plausible explanation (top down and bottom up)
If the premises are true and the reasoning is valid, the conclusion must be true. *It is guaranteed.*	**The conclusion is probable, not certain, even with strong evidence.** It is *provisional, not guaranteed.*	The **conclusion is the best current explanation**, but alternatives may exist. It is *provisional, not guaranteed.*
Direction: General → Specific	**Direction:** Specific → General	**Direction:** Observation → explanation
Used in mathematics, formal logic, and hypothesis testing.	Used in science, statistics, and everyday decision-making.	Used in diagnosis, scientific hypothesis generation, medicine, and troubleshooting

Anatomy of an Argument

Premise: The evidence for making the claim

Warrant: The connection between the premise and the claim

Claim: What you are asserting to be true (the conclusion)

Premise + Warrant → Claim

Anatomy of an Argument: example

Premise: The current protocol misses 30% of cases in the target population

+

Warrant: A protocol that misses 30% of cases is unacceptably inaccurate, and accuracy is the primary criterion for protocol selection

=

Claim: We should change our screening protocol

Premise + Warrant → Claim

Quality of an Argument

Validity:
A valid argument has a conclusion that follows logically from its premises.

Soundness:
A sound argument is **valid** and **has true premises.**

Sound Argument→ Reasoning Valid + True Premises

Quality of an Argument: Sound arguments

Component	Statement
Premise (Evidence)	• This patient tested positive for influenza A using a PCR test with high sensitivity and specificity.
Warrant	• If a highly accurate diagnostic test for influenza A is positive, then it is reasonable to conclude the patient is infected with influenza A (assuming the test was administered correctly).
Claim	• Therefore, this patient has an influenza A infection.

Component	Statement
Premise (Evidence)	• All mammals are warm-blooded • Dolphins are mammals
Warrant	• If something belongs to a category, then it inherits the defining properties of that category
Claim	Therefore, Dolphins are warm-blooded.

Quality of an Argument: Unsound arguments

False premise:

Component	Statement
Premise (Evidence)	• All fish can fly
Warrant	• If something belongs to a category, then it inherits the defining properties of that category
Claim	• Therefore, salmon can fly

Faulty Warrant

Component	Statement
Premise (Evidence)	• Patient has a fever
Warrant	• Anyone with a fever has influenza
Claim	• Therefore, this patient has influenza.

Quality of an Argument

Example	Premise	Warrant	Result
Sound	True	Valid	Trust the claim
Unsound #1	**False**	Valid	Garbage in → garbage out
Unsound #2	True	**Faulty**	Bad reasoning despite good evidence

Strength of Evidence

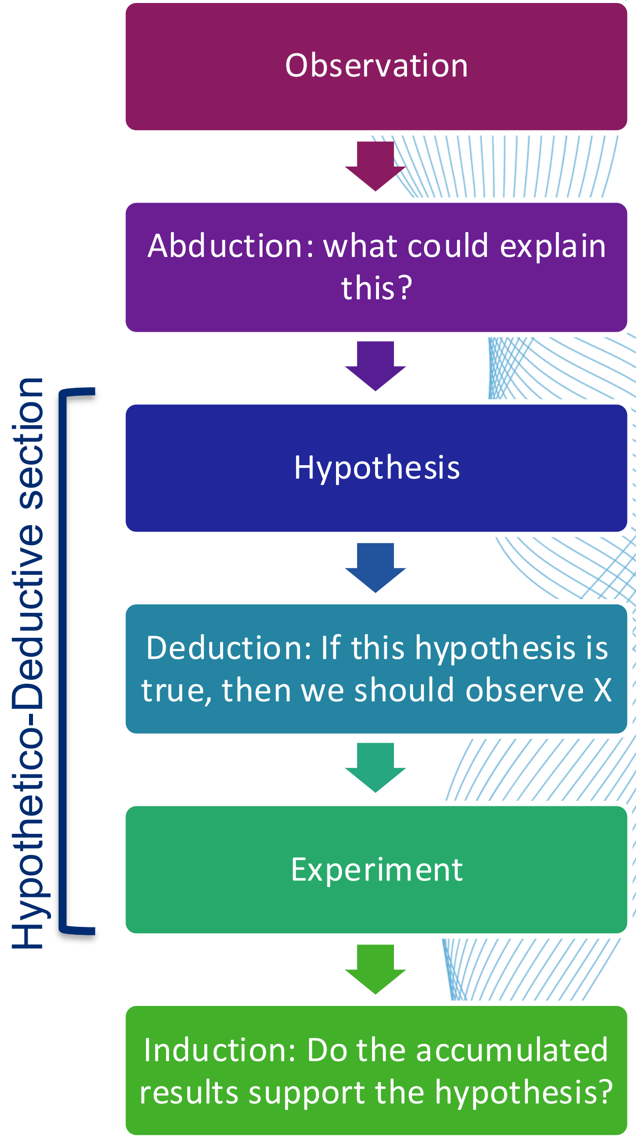
1. In my experience, this approach works
2. Three published studies in comparable populations show this approach works
3. A meta-analysis of 40 trials shows this approach works, though effect sizes vary

Rate the above three premises from 1(weak) to 5 (strong)

Hypothetico-Deductive reasoning

In hypothesis **testing** (which is based on refutation), this is written as:

(1) If H, then E → if H_0 then E is highly probable
(2) Not E → observe data that is unlikely under H_0
(3) Therefore, Not H → reject H_0 (with level of uncertainty)



https://en.wikipedia.org/wiki/Hypothetico-deductive_model

Wason Demonstration

Assertion: *If a card has a vowel on one side, it has an even number on the other side*

What is the minimum number of cards – and which ones - do you need to turn over to test whether this rule is true or false?

E	K	4	7

Wason Demonstration

Which of the following people's ID should be checked to ensure that no-one under 21 drinks alcohol?

16	21	Beer	Soda

A curated list of Common Fallacies

(1) **Ad hominem** – attacking the source rather than the argument; using source credibility as a substitute for engaging with the evidence.

Example: Dr. X has a financial conflict of interest, so her findings are wrong

https://www.fallacyfiles.org/examples.html

A curated list of Common Fallacies

(1) **Ad hominem** – attacking the source rather than the argument; using source credibility as a substitute for engaging with the evidence.
(2) **Appeal to Authority** – accepting a claim because an authority figure made it, without examining the evidence.

Example: The department chair recommended this protocol, so it must be the right one

A curated list of Common Fallacies

(1) **Ad hominem** – attacking the source rather than the argument; using source credibility as a substitute for engaging with the evidence.
(2) **Appeal to Authority** – accepting a claim because an authority figure made it, without examining the evidence.
(3) **False Dichotomy** – presenting two options as exhaustive when other exist.

Example: Either we accept this trial's conclusions, or we abandon evidence-based practice entirely

A curated list of Common Fallacies

(1) **Ad hominem** – attacking the source rather than the argument; using source credibility as a substitute for engaging with the evidence.
(2) **Appeal to Authority** – accepting a claim because an authority figure made it, without examining the evidence.
(3) **False Dichotomy** – presenting two options as exhaustive when other exist.
(4) **Confirmation Bias** – selectively presenting only evidence that supports the claim while ignoring disconfirming evidence.

Example: What evidence against this claim did you look for? What did you find?

Wason Demonstration shows confirmation bias

A curated list of Common Fallacies

(1) **Ad hominem** – attacking the source rather than the argument; using source credibility as a substitute for engaging with the evidence.
(2) **Appeal to Authority** – accepting a claim because an authority figure made it, without examining the evidence.
(3) **False Dichotomy** – presenting two options as exhaustive when other exist.
(4) **Confirmation Bias** – selectively presenting only evidence that supports the claim while ignoring disconfirming evidence.
(5) **Post Hoc Ergo Propter Hoc** – "After this, therefore **because** of this"; confusing temporal sequence with causation

Example: We implemented the new intake process and readmissions dropped; therefore, the process works

A curated list of Common Fallacies

(1) **Ad hominem** – attacking the source rather than the argument; using source credibility as a substitute for engaging with the evidence.
(2) **Appeal to Authority** – accepting a claim because an authority figure made it, without examining the evidence.
(3) **False Dichotomy** – presenting two options as exhaustive when other exist.
(4) **Confirmation Bias** – selectively presenting only evidence that supports the claim while ignoring disconfirming evidence.
(5) **Post Hoc Ergo Propter Hoc** – "After this, therefore **because** of this"; confusing temporal sequence with causation
(6) **Slippery Slope** - asserting that one step will inevitably lead to a chain of consequences, without demonstrating causal links.

Example: If we allow this exception to the protocol, eventually protocols will not mean anything.

Anatomy of an Argument

Premise: The evidence for making the claim

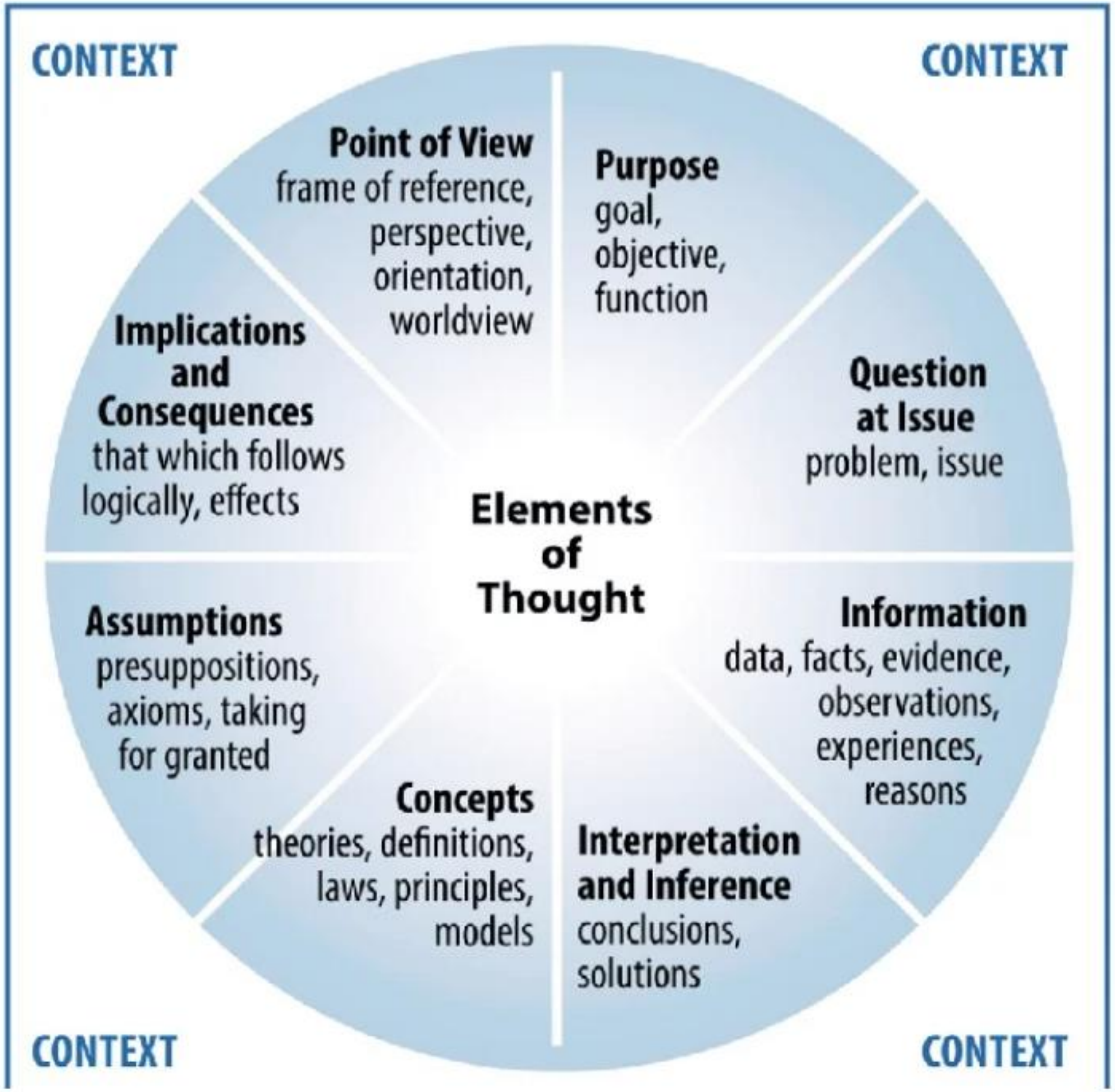
Warrant: The connection between the premise and the claim

Claim: What you are asserting to be true (the conclusion)

Premise + Warrant → Claim

The Paul-Elder Framework

Lecture



Elements of thought, reproduced from Paul & Elder (2019)

Anatomy/components

Evaluation criteria for each Element

Paul-Elder Framework

Elements of Thought

1. Purpose
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Intellectual Standards

1. Clarity
2. Accuracy
3. Precision
4. Relevance
5. Depth
6. Breadth
7. Logic
8. Significance
9. Fairness

Elements x Standards = Analysis

Anatomy/components

Evaluation criteria for each Element

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Intellectual Standards

1. Clarity
2. Accuracy
3. Precision
4. Relevance

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information: What data, evidence, and observations is this based on?
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information: What data, evidence, and observations is this based on?
4. Interpretation & Inference: What conclusions are being drawn from the information?
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information: What data, evidence, and observations is this based on?
4. Interpretation & Inference: *What conclusions are being drawn from the information?*
5. Concepts: What key terms and ideas is this reasoning built on, and are they being used consistently and clearly?
6. Assumptions
7. Implications & Consequences
8. Point of View

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information: What data, evidence, and observations is this based on?
4. Interpretation & Inference: *What conclusions are being drawn from the information?*
5. Concepts: What key terms and ideas is this reasoning built on, and are they being used consistently and clearly?
6. Assumptions: What is being taken for granted that has not been established?
7. Implications & Consequences
8. Point of View

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information: What data, evidence, and observations is this based on?
4. Interpretation & Inference: *What conclusions are being drawn from the information?*
5. Concepts: What key terms and ideas is this reasoning built on, and are they being used consistently and clearly?
6. Assumptions: What is being taken for granted that has not been established?
7. Implications & Consequences: If this reasoning is correct, what follows? What are the downstream effects of accepting this conclusion?
8. Point of View

Elements x Standards = Analysis

Paul-Elder Framework

Elements of Thought

1. Purpose: Why is this claim being made?
2. Question at Issue: What specific question is this reasoning trying to answer?
3. Information: What data, evidence, and observations is this based on?
4. Interpretation & Inference: *What conclusions are being drawn from the information?*
5. Concepts: What key terms and ideas is this reasoning built on, and are they being used consistently and clearly?
6. Assumptions: What is being taken for granted that has not been established?
7. Implications & Consequences: If this reasoning is correct, what follows? What are the downstream effects of accepting this conclusion?
8. Point of View: From what perspective is this reasoning being conducted? What perspectives are absent?

Elements x Standards = Analysis

Anatomy/components

Evaluation criteria for each Element

Paul-Elder Framework

Elements of Thought

1. Purpose
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Intellectual Standards

1. Clarity
2. Accuracy
3. Precision
4. Relevance
5. Depth
6. Breadth
7. Logic
8. Significance
9. Fairness

Elements x Standards = Analysis

Intellectual Autopsy

Activity 1

Paul-Elder Framework

Given the incentives, tools, and uncertainty, what would *you* have done differently, and at what cost?

How do you operate in a system where flawed evidence can look legitimate and be rewarded?

> Nature. 2006 Mar 16;440(7082):352-7. doi: 10.1038/nature04533.

A specific amyloid-beta protein assembly in the brain impairs memory

Sylvain Lesné [1], Ming Teng Koh, Linda Kotilinek, Rakez Kayed, Charles G Glabe, Austin Yang, Michela Gallagher, Karen H Ashe

Affiliations + expand

PMID: 16541076 DOI: 10.1038/nature04533

Retraction in

Retraction Note: A specific amyloid-β protein assembly in the brain impairs memory.

Lesné S, Koh MT, Kotilinek L, Kayed R, Glabe CG, Yang A, Gallagher M, Ashe KH.

Nature. 2024 Jul;631(8019):240. doi: 10.1038/s41586-024-07691-8.

PMID: 38914864 No abstract available.

Abstract

Memory function often declines with age, and is believed to deteriorate initially because of changes in synaptic function rather than loss of neurons. Some individuals then go on to develop Alzheimer's disease with neurodegeneration. Here we use Tg2576 mice, which express a human amyloid-beta precursor protein (APP) variant linked to Alzheimer's disease, to investigate the cause of memory decline in the absence of neurodegeneration or amyloid-beta protein amyloidosis. Young Tg2576 mice (< 6 months old) have normal memory and lack neuropathology, middle-aged mice (6-14 months old) develop memory deficits without neuronal loss, and old mice (> 14 months old) form abundant neuritic plaques containing amyloid-beta (refs 3-6). We found that memory deficits in middle-aged Tg2576 mice are caused by the extracellular accumulation of a 56-kDa soluble amyloid-beta assembly, which we term Abeta*56 (Abeta star 56). Abeta*56 purified from the brains of impaired Tg2576 mice disrupts memory when administered to young rats. We propose that Abeta*56 impairs memory independently of plaques or neuronal loss, and may contribute to cognitive deficits associated with Alzheimer's disease.

Paul-Elder Framework Autopsy

In 2006, a research team published a paper in Nature claiming that a specific protein assembly called ***Aβ*****56** caused memory impairment in mice, a finding that pointed toward a promising Alzheimer's therapeutic target.

The paper was cited over 2,300 times and influenced drug development for nearly two decades.

In 2022, a forensic scientist examining the images in this and related papers concluded that the Western blot images, the visual evidence for Aβ*56's existence, appeared to have been digitally altered.*

The paper was retracted in 2024.

It is now the second most-cited retraction in scientific history.

Paul-Elder Framework

- Besides failures that are identified by the Paul-Elder Framework there is also

incentive failure & Motivated Reasoning

- Useful Question: What incentives would produce this behavior even if nobody intended harm?

“It is difficult to get a man to understand something, when his salary depends upon his not understanding it” Upton Sinclair, 1934

Paul-Elder Framework Autopsy Instructions

- ***Round 1 (12 min):*** *Working in pairs, map the Lesné paper to Paul-Elder's 8 elements. For each element, write one sentence describing what is present in the paper, and one sentence noting what is absent or weak.*
- ***Round 2 (8 min):*** *Choose the two elements your pair considers weakest. Apply the four standards (clarity, accuracy, precision, relevance) to those two elements only.*
- ***Debrief (10 min):*** *Full group. Three questions:*
  1. *Which element did your pair identify as the most significant failure?*
  2. *Which standard did this paper most clearly violate?*
  3. ***What would it have cost to apply these standards at the time of publication?***

Paul-Elder Framework

	Clarity	Accuracy	Precision	Relevance
Purpose				
Question				
Information				
Interpretation				
Concepts				
Assumptions				
Consequences				
POV				

Elements x Standards = Analysis

Unpack Your Brain

Activity 2

Unpack your brain

You just applied Paul-Elder Framework to someone else's reasoning; now you have 8 minutes to apply it to your own

One conclusion that you hold strongly in your current work?

- *What is your purpose, and is it the purpose you are claiming?*
- *What question are you actually answering, and is it the right one?*
- *What information are you relying on, and what have you not looked at?*
- *What are your assumptions, stated and unstated?*
- *Whose point of view is absent from your reasoning?*

Polya’s 4 Step method

Lecture

Polya's 4 steps

UNDERSTANDING THE PROBLEM

First.

You have to *understand* the problem.

What is the unknown? What are the data? What is the condition? Is it possible to satisfy the condition? Is the condition sufficient to determine the unknown? Or is it insufficient? Or redundant? Or contradictory?

Draw a figure. Introduce suitable notation.

Separate the various parts of the condition. Can you write them down?

DEVISING A PLAN

Second.

Find the connection between the data and the unknown. You may be obliged to consider auxiliary problems if an immediate connection cannot be found. You should obtain eventually a *plan* of the solution.

Have you seen it before? Or have you seen the same problem in a slightly different form?

Do you know a related problem? Do you know a theorem that could be useful?

Look at the unknown! And try to think of a familiar problem having the same or a similar unknown.

Here is a problem related to yours and solved before. Could you use it? Could you use its result? Could you use its method? Should you introduce some auxiliary element in order to make its use possible?

Could you restate the problem? *Could you restate it still differently?* Go back to definitions.

If you cannot solve the proposed problem try to solve first some related problem. Could you imagine a more accessible related problem? A more general problem? A more special problem? An analogous problem? Could you solve a part of the problem? Keep only a part of the condition, drop the other part; how far is the unknown then determined, how can it vary? Could you derive something useful from the data? Could you think of other data appropriate to determine the unknown? Could you change the unknown or the data, or both if necessary, so that the new unknown and the new data are nearer to each other?

Did you use all the data? Did you use the whole condition? Have you taken into account all essential notions involved in the problem?

CARRYING OUT THE PLAN

Third.

Carry out your plan.

Carrying out your plan of the solution, *check each step*. Can you see clearly that the step is correct? Can you prove that it is correct?

LOOKING BACK

Fourth.

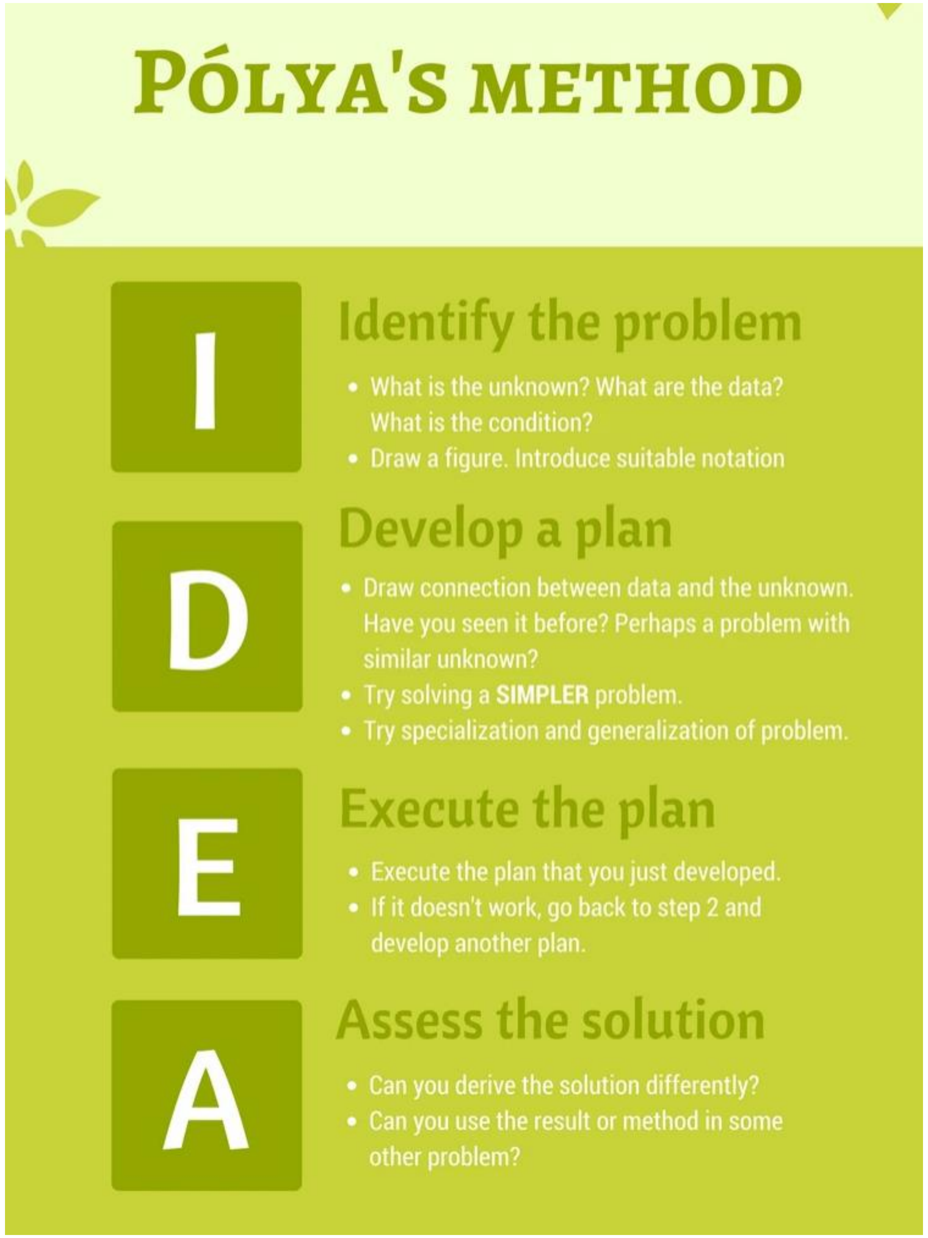
Examine the solution obtained.

Can you *check the result?* Can you check the argument?

Can you derive the result differently? Can you see it at a glance?

Can you use the result, or the method, for some other problem?

Polya’s 4 steps



Fermi Estimation (using Polya)

How many MRI machines are currently operating in the United States?

Step 1: Understand the problem

Step 2: Devise a plan

Step 3: Execute the plan

Step 4: Look back

Fermi Estimation (using Polya) De-Brief

- Each group will walk through their answer
- **Which step produced the most variation?**
- **What is one assumption that most affected your estimate?**

The Four Frameworks

Kahneman

System 1 - *FAST*
- Uses heuristics

System 2 – *SLOW*
- Deliberate, and analytical
- Cognitively demanding

Paul-Elder

1. Purpose
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Polya

1. Understand Problem
2. Devise a Plan
3. Execute Plan
4. Iteration & Reflection

Wild & Pfannkuch

1. Problem
2. Plan
3. Data
4. Analysis
5. Conclusion

Calibration failures these model's address:

We are unaware that fast thinking is operating at all!

We argue from hidden assumptions that we never examine!

We treat complex problems as unsolvable wholes!

We trust data that encodes errors that we cannot see!

The four frameworks

Framework	Where to Focus	Key Failures / Breakdowns	Core Insight
Kahneman (System 1 / System 2)	Collective cognition under uncertainty	**System 1:** narrative coherence ("amyloid causes Alzheimer's"), hope bias, authority heuristic, citation cascade **System 2:** insufficient skepticism of clean data, weak interrogation of reproducibility, high cost of dissent	System 2 doesn't fail just from laziness; it fails when scrutiny is costly (socially, financially, professionally)
Paul–Elder (Structure of Reasoning)	Deconstruct the argument and its assumptions	**Purpose:** mixed (truth vs publication/funding) **Question:** prematurely narrowed **Assumptions:** amyloid causality, model validity **Evidence:** potentially manipulated / selectively interpreted **Inferences:** correlation → causation leap **Point of View:** field-level investment **Standards violated:** accuracy, breadth, logic	Most failures are not obviously illogical; they are incomplete, narrow, or overextended
Polya (Problem-Solving Process)	How the problem was defined and pursued	**Understand:** oversimplified disease complexity **Plan:** narrow mechanistic focus, weak alternatives **Execute:** possible misconduct or biased interpretation **Look Back:** weak replication culture, delayed skepticism	"Look back" is the most neglected and most critical step in real-world problem solving
PPDAC (Statistical Cycle)	Full data lifecycle (not just analysis)	**Problem:** too narrowly framed **Plan:** fragile experimental design choices **Data:** potential manipulation, lack of transparency **Analysis:** overinterpretation, noise ignored **Conclusion:** strong causal claims from weak base	Data does not fail in one place; entire pipelines can produce convincing but misleading conclusions

Homework

Solving Problems: Day 2 – Fill out form